

School of Arts and Sciences

Department of Computer Science and Information Technology

Fall 2025 - 2026

CSCI205 Computer Science Overview

Class Day	:	_____	Class Time:	_____	Room:	_____
Instructor	:	_____				
E-mail	:	_____				
Office	:	_____	Office Hours:	_____	Extension No:	_____

I. Course Description

This course presents breadth coverage of computer science courses so that students would understand computing and appreciate technology's impact on society. Topics include binary values and number systems; data representation; gates and circuits; computing components; operating systems; file systems and directories; information systems; computer networks; Flowcharting & Algorithms; and elementary Programming.

II. Course Objectives

CO-1: Explain how data is represented, manipulated and stored within a computer system.

CO-2: Identify and explain the functions of operating systems.

CO-3: Explain how computers interact through local and wide area networks.

CO-4: Build Flowcharts, write algorithms and explain common data types and programming structures.

CO-5: Explore the Computer Science major.

III. Course Learning Outcomes

CLO-1: Describe different methods of data representation and storage within the computer system as well as data manipulation in binary, decimal, and hexadecimal number systems.

CLO-2: Recognize and use machine language.

CLO-3: Identify the main functionalities and the leading architecture of operating systems, CPU, and networks.

CLO-4: Explain the advantages of the database over the traditional files system and the main data manipulation functions.

CLO-5: Convert flowchart and algorithms to adequately structured and commented java source codes.

IV. Mapping to program outcomes (*PLOs are listed at the end of the syllabus*)

Computer Science Overview	PLO -1	PLO -2	PLO -3	PLO -4	PLO -5	PLO -6	PLO -7	PLO -8	PLO -9
CLO-1	X								
CLO-2	X								
CLO-3	X								
CLO-4		X							
CLO-5				X					

V. Topics Covered

- 1 Data Storage
- 2 Data Manipulation
- 3 Operating Systems
- 4 Networking and the Internet
- 5 Database Systems
- 6 Flowcharting and Algorithms
- 7 Introduction to Java
- 8 Elementary Programming
- 9 Selections

VI. Prerequisite – Corequisite

ENGL101 (prerequisite)

CSCI200 (Corequisite)

VII. Course Material:

Book 1 (Computer Science - An Overview 12th edition: Glenn Brookshear & Dennis Brylow, **ISBN-10:** 0133760065, Pearson, 2015)

Book 2 (Introduction to Java Programming 9th edition: Daniel Liang, **ISBN-10:** 0132936526, Pearson, 2013)

Book 3 (Flowchart and Algorithm Basics 1st edition: A.B. Chaudhuri, **ISBN:** 978-1-68392-537-8, Mercury Learning and Information LLC, 2020)
or handouts

VIII. Software(s) required

NetBeans

IX. Course Assessment

# Activity	Percentage
Test-1 (non-common)	15
Midterm	30
Test-2 (non-common)	20
Final Exam	35

X. Detailed Schedule

Week		Book Reference	Topic Details
Week 1 (Sep 29 – Oct 2)	Lecture 1	Computer Science - An Overview	Syllabus: reading and explaining. Chapter 1: Data Storage 1.1 Bits and Their Storage (Boolean operations, and gates)
	Lecture 2	Computer Science - An Overview	Chapter 1: Data Storage 1.1 Bits and Their Storage (hexadecimal notation) 1.2 Main Memory (memory organization, measuring memory capacity) 1.3 Mass Storage (magnetic systems, optical systems, flash drives, file storage and retrieval - in brief)
Week 2 (Oct 5 - 9)	Lecture 1	Computer Science - An Overview	Chapter 1: Data Storage 1.4 Representing Information as Bit Patterns (representing text, representing numeric values) 1.5 The Binary System (binary notation, binary addition, fractions in binary)
	Lecture 2	Computer Science - An Overview	Chapter 1: Data Storage Solve Chapter Review Problems 1, 2, 7, 19, 26, & 27 Pages 85-87.
Week 3 (Oct 13 - 16)	Lecture 1	Computer Science - An Overview	Chapter 1: Data Storage 1.6 Storing Integers (two's complement notation, overflow problem) Solve Chapter Review Problem 30 Page 87.
	Lecture 2	Computer Science - An Overview	Chapter 1: Data Storage 1.7 Storing Fractions (Excess Notation, floating-point notation, truncation errors) Solve Chapter Review Problems 32, 34 and 36 Page 87.
Week 4 (Oct 20 - 23)	Lecture 1	Computer Science - An Overview	Chapter 1: Data Storage Solve Chapter Review Problems 31, 33, 35, and 46 Page 87-88.
	Lecture 2	Computer Science - An Overview	Chapter 2: Data Manipulation 2.1 Computer Architecture (CPU basics, the stored-program concept) 2.2 Machine Language (the instruction repertoire)
Week 5 (Oct 27 - 30)	Lecture 1	Instructor's handouts	Non-Common Test 1 covering Chapter 1 - done during class time
	Lecture 2	Computer Science - An Overview	Chapter 2: Data Manipulation 2.2 Machine Language (an illustrative machine language) 2.3 Program Execution (an example of program execution, program versus data) Solve instructor exercises
Week 6 (Nov 3 - 6)	Lecture 1	Computer Science - An Overview	Chapter 2: Data Manipulation Solve Chapter Review Problems 5, 12, 13, 14, 16, and 22 Pages 131-133
	Lecture 2	Instructor's handouts	Chapter 2: Data Manipulation Solve instructor exercises.
Week 7 (Nov 10 - 13)	Lecture 1	Computer Science - An Overview	Chapter 3: Operating Systems 3.1 The History of Operating Systems 3.2 Operating System Architecture (a software survey, components of an operating system, getting it started)
	Lecture 2	Computer Science - An Overview	Chapter 4: Networking and internet 4.1 Network Fundamentals (network classifications, protocols, combining networks, methods of process communication) 4.2 The Internet (internet architecture, internet addressing, internet applications - brief)
Week 8 (Nov 17 - 20)	Lecture 1	Computer Science - An Overview	Chapter 9: Database Systems 9.1 Database Fundamentals (the significance of database systems, the role of schemas, Database Management Systems, Database models) 9.2 The Relational Model (issues of relational design) (all in brief)

	Lecture 2	Computer Science - An Overview	Chapter 9: Database Systems 9.2 The Relational Model (Relational Operations and SQL)
Midterm Exam – Covering Chapters 1, 2, 3 and 4			
Week 9 (Nov 24 - 27)	Lecture 1	Computer Science - An Overview Instructor's handouts	Chapter 9: Database Systems Solve Chapter Review Problems 12, 13 & 15 Page 451, 452 and Instructor's additional problems
	Lecture 2	Flowchart and Algorithm Basics	Chapter 1: Introduction to Programming, Flowcharting and Algorithms, solve problems 1.1, 1.2, 1.3, and 1.4 Assign as assignment the exercises: 1.5 till 1.11
Week 10 (Dec 1 - 4)	Lecture 1	Flowchart and Algorithm Basics	Chapter 2: Problems Involving Selection, Introduction, solve problems 2.2, 2.3, 2.4, 2.5, and 2.11
	Lecture 2	Flowchart and Algorithm Basics	Chapter 2: Problems Involving Selection, and solve instructor exercises for test 2.
Week 11 (Dec 8 – 11)	Lecture 1	Instructor's handouts	<i>Non-Common Test 2 covering Database, Flowchart & Alg. chapters 1 & 2</i>
	Lecture 2	Introduction to Java Programming	Chapter 1: Introduction to Computers Programs and Java 1.3. Programming Languages 1.6 The Java Language Specification, API, JDK, and IDE 1.7 A Simple Java Program 1.11 Programming Errors (Syntax, Runtime, Logic Errors)
Week 12 (Dec 15 - 18)	Lecture 1	Introduction to Java Programming	Chapter 2: Elementary Programming 2.4 Identifiers 2.5 Variables 2.3 Reading Input from the Console (including scan.next().charAt(0)) 2.2 Writing a Simple Program
	Lecture 2	Introduction to Java Programming	Chapter 2: Elementary Programming 2.6 Assignment Statements and Assignment Expressions 2.7 Named Constants
Week 13 (Dec 22 - 25)	Lecture 1	Introduction to Java Programming	<i>December 24 and 25 - Christmas Vacation</i> Chapter 2: Elementary Programming 2.9 Numeric Data Types and Operations 2.9.1 Numeric Types 2.9.2 Numeric Operators 2.10 Numeric Literals (integers vs. floating-point)
Week 14 (Jan 5 - 8)	Lecture 1	Introduction to Java Programming	<i>January 6 – Armenian Christmas</i> Chapter 2: Elementary Programming Solve Problems 2.2, 2.4, 2.6, and 2.12, Pages 75-77 <i>TTH colleagues share the problems' solutions on GCR</i>
	Lecture 2	Instructor's handouts	Review for the final exam

XI. Class Rules & Discipline

Attendance Policy

The student is administratively withdrawn (AW) from the class if he/she is absent for more than one third of the classes (15 class sessions for courses offered 3 times a week, 10 for courses offered twice a week, and 5 for courses offered once a week). If the number of absences exceeds the limit after the withdrawal deadline, i.e. including all the period up-to the end of the semester, the student will not receive an AW grade, therefore, the final examination must be written, otherwise, an F grade will be granted. When absent, you are responsible for all material presented in class. Missed exams will only be excused by documented evidence as explained in the make-up examination policy below.

A “10%” has been allocated on the “Marking Profile” for attendance. A student who hasn’t missed any lecture will get a full mark on the attendance grade. Every absence, a 20% will be deducted, that is, a student with 5 absences will get a ZERO on that grade.

Cheating

The University policy on cheating will be strictly applied. Students are required to adhere to the policy on academic irregularities contained in the LIU student handbook. Collaboration among students in solving assignments is not acceptable. It is okay to ask another student for advice when stuck on a problem, but if the instructor receives assignments that are substantially equivalent, all involved students will be punished with a grade of zero.

Classroom Manners

All students are expected to conduct themselves in a professional manner. Students should not: keep their cell phones on during the class, disrupt the class, do other course work in class, have side conversations with other students, be rude towards anyone. The first failure to respect these rules will result in a *warning* from the instructor. If a student chooses to disregard the *warning*, he/she will be dismissed from class.

Instructor late to the class

In case the instructor is late the students should wait for 15 minutes before leaving the class.

Abuse of Computing Privileges

The Office of Information Technology maintains policies posted in the computer labs regarding proper behavior in the labs. Failure to adhere to these policies may result in loss of computer privileges, and possible legal action.

Make-Up Examination Policy

As per university council decision, a student is eligible for a mid-term or final examination make-up if and only if he/she had the following incidents:

- a. Sickness; **proved by hospitalization report**; that is; a discharge summary is necessary.
- b. Death in the family **proved by a death certificate** and personal identification.
- c. Accidents **proved by an expert report**.

A student that misses an exam should submit a petition within one week from the examination date. Late petitions will be rejected unless it is an extreme case (Hospitalization as an example).

In case a petition is accepted, the student submits a MAKE-UP exam. The grade of the missing exam will be equal to a percentage % of the Final Exam’s grade. The percentage % will be set by the administration once the petition is approved.

Cheating policy: Any cheating will be rewarded with a zero, all students involved in cheating will get a zero (even if the cheating is in one question). The activity would not be repeated.

Missing Assignments and Projects Policy

- A -10 per day penalty will be applied to late projects (weekends count as two days).
- A ZERO will be applied for late assignments on the next day.

Curriculum, Yearly Plan, and Advising

If a student majoring in Computer Science or Information Technology is planning to graduate in three years then he/she should follow the corresponding major yearly plan. A copy of the yearly plan and the curriculum (which contains the prerequisites and co-requisites of courses) can be picked up from the office of the school of Arts and Science. When not sure about anything please contact your advisor prior to or during the registration period.

Probation Policy

If a student who has attempted 19 credits has a GPA or a major GPA below 2.0 then the student is considered under probation.

mails

Students should use their university e-mails in communicating with the instructors. Important announcements, syllabi, assignments, and solutions will be sent to your LIU e-mail accounts.

University Peer Review Policy

All classes will be visited by instructors either from the same or different departments. This policy helps instructors figure out ways to improve student educational experiences in class and eventually their performance.

Snapshots

Students are not allowed to take snapshots of the whiteboard in classrooms using their mobile-phones.

Equivalent Lettering Grades

<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>	<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>
<60	0	F	67	1.7	C
60	1	D	68	1.8	
61	1.1		69	1.9	
62	1.2		70	2	
63	1.3		71	2.1	
64	1.4		72	2.2	
65	1.5	D+	73	2.3	B+
66	1.6		74	2.4	
75	2.5	C+	83	3.3	
76	2.6		84	3.4	
77	2.7		85	3.5	
78	2.8		86	3.6	
79	2.9		87	3.7	
80	3	B	88	3.8	A
81	3.1		89	3.9	
82	3.2		>=90	4	

<u>Grade</u>	<u>Description</u>	<u>QPTS</u>
A	Excellent	4.00
B+	Very Good	3.50
B	Good	3.00
C+	Above Average	2.50
C	Average	2.00
D+	Fair	1.50
D	Poor	1.00
F	Fail	0.00
IP	Course in Progress	
I	Incomplete work	
W	Student has withdrawn from the course	
AW	Administrative withdrawal from the course	
R	Course has been repeated	
S	Satisfactory work	
U	Unsatisfactory work	
%G	Final Grade	
QPTS	Quality Points	

Computer Science Outcomes

Graduates of the Computer Science Program will have the ability to:

	Program Outcomes
category -1 Underlying Conceptual Basis for Informatics	PLO -1 Describe and explain the essential facts, concepts, theories and mathematical methods relevant to computing, computing equipment, computer communication and informatics applications as appropriate to their program of study
	<i>PLO-1 for CSIT=> Gain basic skills in mathematics and outline the characteristics of relevant state-of-the-art hardware and software and their practical application</i>
category -2 Analysis	PLO -2 Select and use relevant analytic, modelling and simulation methods
	PLO-3 Choose appropriate solution patterns, algorithms and data structures
category -3 Design and Implementation	PLO -4 Describe the software development process and apply relevant practical and programming skills to the creation of computer programs, and/or other informatics artefacts
	<i>PLO-4 for CSIT=>Specify and design computing/network hardware/software which meet specified requirements</i>
category -4 Economic, legal, social, ethical and environmental context	PLO -5 Explain the importance of information privacy and security issues in relation to the design, development, maintenance, monitoring and use of informatics-based systems
category -5 Informatics practice	PLO -6 Describe and explain management techniques appropriate to the design, implementation, testing, deployment and maintenance of informatics systems, including project management, configuration management, change management, etc., and including relevant automated techniques
	<i>PLO -6 for CSIT=> Demonstrate an awareness of project management and business practices, such as risk and change management, and understand their limitations;</i>
category -6 Other Professional Skills and Competences	PLO -7 Communicate effectively both verbally and using a variety of communications media
	PLO -8 Plan self-learning and improve personal performance as a foundation for lifelong learning and ongoing professional development
	PLO -9 Participate effectively in informatics group-working